Algebra-0 2026 Spring

Homework 6

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*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

$\boxed{\textbf{Problem 1 } \text{(5A-29).}}$ Suppose $T \in \mathcal{L}(\mathbb{R}^3)$ and -4, 5, and $\sqrt{7}$ are eigenvalues of T. Prove that there exists $x \in \mathbb{R}^3$ such that

$$Tx - 9x = (-4, 5, \sqrt{7}).$$

Proof. Since -4, 5, and $\sqrt{7}$ are eigenvalues of $T \in \mathcal{L}(\mathbb{R}^3)$, we can infer that 9 is not an eigenvalue, hence $\ker(T - 9I) = \{0\}$. Therefore, $T - 9I$ is injective. Since $\mathbb{R}^3$ is finite-dimensional, injectivity implies surjectivity. Thus, for any vector v in $\mathbb{R}^3$, there exists a vector $x \in \mathbb{R}^3$ such that $Tx - 9x = v$. Take $v = (-4, 5, \sqrt{7})$, then the problem has been proved. □

$\boxed{\textbf{Problem 2 } \text{(5A-30).}}$ Suppose $T \in \mathcal{L}(V)$ and

$$(T - 2I)(T - 3I)(T - 4I) = 0.$$

Suppose λ is an eigenvalue of T. Prove that $\lambda = 2$ or $\lambda = 3$ or $\lambda = 4$.

Proof. Take $v \neq 0$ such that $Tv = \lambda v$. Then

$$0 = (T - 2I)(T - 3I)(T - 4I)v = (\lambda - 2)(\lambda - 3)(\lambda - 4)v \implies \lambda \in \{2, 3, 4\}.$$

□

$\boxed{\textbf{Problem 3 } \text{(5A-38).}}$ Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V invariant under T. The quotient operator $T/U \in \mathcal{L}(V/U)$ is defined by

$$(T/U)(v + U) = Tv + U$$

for each $v \in V$.

(a) Show that the definition of T/U makes sense (which requires using the condition that U is invariant under T) and show that T/U is an operator on V/U.

(b) Show that each eigenvalue of T/U is an eigenvalue of T.

Proof. (a) For any $v_1, v_2 \in V$ such that $v_1 + U = v_2 + U$, we know that $u := v_1 - v_2 \in U$. Then

$$\begin{aligned}(T/U)(v_1 + U) &= T(v_1) + U = T(v_2 + u) + U = T(v_2) + Tu + U \\ &= T(v_2) + U = (T/U)(v_2 + U).\end{aligned}$$

Then T/U is well-defined.

(b) Take an eigenvalue and corresponding eigenvector of T/U, say λ and $v+U$. Note that $v+U \neq U$, so $v \notin U$. Then $Tv + U = (T/U)(v+U) = \lambda(v+U) = \lambda v + U$, which implies that $Tv = \lambda v + u$ for some $u \in U$.

If λ is not an eigenvalue of T, then $T - \lambda I$ is injective in V, as well as in U (because U is T-invariant). Then there exists $u_0 \in U$ such that $Tu_0 - \lambda u_0 = u \in U$.

Set $v' = v - u_0$. Since $v \notin U$, we have $v' \neq 0$. And by the definition of u_0, we have

$$Tv' = Tv - Tu_0 = \lambda v + u - (\lambda u_0 + u) = \lambda(v - u_0) = \lambda v'.$$

This contradicts the assumption that λ is not an eigenvalue of T. Therefore, λ is an eigenvalue of T. □

Problem 4 (5A-42). Define $T \in \mathcal{L}(\mathbb{F}^n)$ by

$$T(x_1, x_2, x_3, \ldots, x_n) = (x_1, 2x_2, 3x_3, \ldots, nx_n).$$

(a) Find all eigenvalues and eigenvectors of T.

(b) Find all subspaces of $\mathbb{F}^n$ that are invariant under T.

Solution. (a) The representation of T under the standard basis is $\operatorname{diag}(1, 2, \ldots, n)$, then all eigenvalues and eigenvectors of T are

$$k \text{ and } (0, \ldots, 0, \overset{(k \text{ th})}{1}, 0, \ldots, 0), \quad k = 1, 2, \ldots, n.$$

(b) We will prove that the invariant subspaces of T are exactly those spanned by a subset of the standard basis, i.e. those of the form

$$V_S := \{a_1, \ldots, a_n \in \mathbb{F}^n : a_i = 0 \text{ for all } i \notin S\}, \text{ where } S \subseteq \{1, 2, \ldots, n\}.$$

Obviously, V_S is invariant under T for any $S \subseteq \{1, 2, \ldots, n\}$.

Conversely, take an invariant subspace U of T. For any $v = (a_1, a_2, \ldots, a_n)$, Define

$$N(v) = \{k \in \{1, 2, \ldots, n\} : a_k \neq 0\},$$

$$v_- = \min\{|a_i| : i \in N(v)\}, \quad v_+ = \max\{|a_i| : i \in N(v)\}.$$

Then, let $S = \bigcup_{u \in U} N(u)$. We will show that $U = V_S$. Obviously $U \subset V_S$, so it suffices to prove that $V_S \subset U$.

First, we will show that there exists $x \in U$ such that $N(x) = S$. Denote S by $S := \{s_1 < \cdots < s_m\}$. For each s_i, there exists $u_i \in U$ such that $s_i \in N(u_i)$. We will construct x inductively by following steps:

- Take $x_1 = u_1$. Then the s_1-th component of x_1 is nonzero.

- Once we have chosen x_k such that s_i-th component of x_k is nonzero for all $1 \le i \le k$. If the s_{k+1}-th component of x_k is nonzero, then take $x_{k+1} = x_k$; otherwise let

$$x_{k+1} = x_k + \frac{(x_k)_-}{2(u_{k+1})_+} u_{k+1}.$$

  The s_{k+1}-th component of x_{k+1} is nonzero due to the choice of u_{k+1}; for any $s_{i(\le k)}$-th component of x_{k+1}, we have

$$\left|x_{k+1}^{(s_i)}\right| = \left|x_k^{(s_i)} + \frac{(x_k)_-}{2(u_{k+1})_+} u_{k+1}^{s_i}\right| \ge \left|x_k^{(s_i)}\right| - \frac{(x_k)_-}{2(u_{k+1})_+}\left|u_{k+1}^{(s_i)}\right| \ge \left|x_k^{(s_i)}\right| - \frac{1}{2}(x_k)_- > 0.$$

  Then the s_i-th component of x_{k+1} is nonzero.

  Hence, we have constructed x_{k+1} such that the s_i-th $(1 \le i \le k+1)$ components are nonzero.

- Finally, we have $x = x_m$ such that $N(x) = S$.

Next, we will show that $V_S \subset \operatorname{span}\{x, Tx, \dots, T^{m-1}x\} \subset U$.

Denote $a_i := x^{(s_i)}$ for any $1 \le i \le m$. Then, for any $v \in V$ $(v^{(s_i)} = b_i)$, the equation

$$\begin{cases}
\lambda_1 a_1 & +\lambda_2 a_1 & +\lambda_3 a_1 & +\dots & +\lambda_m a_1 & = b_1, \\
\lambda_1 a_2 & +\lambda_2 2 a_2 & +\lambda_3 2^2 a_2 & +\dots & +\lambda_m 2^{m-1} a_2 & = b_2, \\
\lambda_1 a_3 & +\lambda_2 3 a_3 & +\lambda_3 3^2 a_3 & +\dots & +\lambda_m 3^{m-1} a_3 & = b_3, \\
\dots \\
\lambda_1 a_m & +\lambda_2 m a_m & +\lambda_m m^2 a_m & +\dots & +\lambda_m m^{m-1} a_m & = b_m
\end{cases}$$

has a solution $(\lambda_1, \dots, \lambda_m)$ because the coefficient matrix is a Vandermonde matrix. Hence,

$$v = \sum_{i=1}^m \lambda_i T^{i-1} x \in \operatorname{span}\left\{x, Tx, \dots, T^{m-1}x\right\} \subset U.$$

Therefore, $V_S \subset U$.

In conclusion, the invariant subspaces of T are exactly those of the form V_S for some $S \subseteq \{1, 2, \dots, n\}$.

Problem 5 (5A-43). Suppose that V is finite-dimensional, $\dim V > 1$, and $T \in \mathcal{L}(V)$. Prove that

$$\{p(T) : p \in \mathcal{P}(\mathbb{F})\} \ne \mathcal{L}(V).$$

Proof. Denote $P = \{p(T) : p \in \mathcal{P}(\mathbb{F})\} \ne \mathcal{L}(V)$. It's easy to verify that P is a subspace of $\mathcal{L}(V)$.

Denote $d := \deg \min_T(x) \le n$. We will show that $\dim P = d < n^2 = \dim \mathcal{L}(V)$, which implies $P \ne \mathcal{L}(V)$.

First, $I, T, T^2, \dots, T^{d-1}$ is linearly independent in $\mathcal{L}(V)$, otherwise there exists not-all-zero $a_0, a_1, \dots, a_{d-1} \in \mathbb{F}$ such that

$$\sum_{k=0}^{d-1} a_k T^k = 0 \implies \min_T(x) \,\Bigg|\, \sum_{k=0}^{d-1} a_k x^k \implies a_k = 0 \text{ for all } 0 \le k \le d-1,$$

which contradicts the choice of $a_0, a_1, \dots, a_{d-1}$. Hence, $\dim P \ge d$.

Next, we will show that $I, T, T^2, \ldots, T^{d-1}$ spans P. For any $p(T) \in P$, we can write $p(x) = q(x)\min_T(x) + r(x)$ for some $q, r \in \mathcal{P}(\mathbb{F})$ such that $r = 0$ or $\deg r < d$. Then

$$p(T) = q(T)\min{}_T(T) + r(T) = r(T) \in \operatorname{span}\left\{I, T, T^2, \ldots, T^{d-1}\right\}.$$

Hence, $\dim P \leq d$.

In conclusion, $\dim P = d < n^2 = \dim \mathcal{L}(V)$, which implies $P \neq \mathcal{L}(V)$. $\square$

Problem 6 (5B-7).

(a) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^2)$ such that the minimal polynomial of ST does not equal the minimal polynomial of TS.

(b) Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that if at least one of S, T is invertible, then the minimal polynomial of ST equals the minimal polynomial of TS.

Solution. (a) Let $S = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $ST = 0$ and $TS = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Hence, the minimal polynomial of ST is x, while the minimal polynomial of TS is $x(x-1)$.

(b) If S is invertible, then $ST = S(TS)S^{-1}$, which implies that ST and TS are similar. Hence, they have the same minimal polynomial. The case when T is invertible can be proved similarly.

Problem 7 (5B-16). Suppose $a_0, \ldots, a_{n-1} \in \mathbb{F}$. Let T be the operator on $\mathbb{F}^n$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & 0 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & -a_{n-2} \\ 0 & 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

Here all entries of the matrix are 0 except for all 1's on the line under the diagonal and the entries in the last column (some of which might also be 0). Show that the minimal polynomial of T is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

Proof. Take $v_k = (0, \ldots, 0, \overset{k\text{-th}}{1}, 0, \ldots, 0)^{\mathrm{T}}$ where the 1 is in the k-th position. Then from the representation matrix of T, we can infer that

$$Tv_1 = v_2,\ Tv_2 = v_3,\ \ldots,\ Tv_{n-1} = v_n,\ Tv_n = -a_0 v_1 - a_1 v_2 - \cdots - a_{n-1} v_{n-1}$$

Denote the minimal polynomial of T by $m(x)$. If $\deg m < n$, then $m(T)(v_1)$ is a linear comnbinition of $v_1, \ldots, v_n$ with not-all-zero coefficients, which contradicts the definition of m. Hence, $\deg m \geq n$. On the other hand, it's easy to verify that

$$T^n v_1 = -a_0 v_1 - a_1 v_2 - \cdots - a_{n-1} v_{n-1} \implies \left(\sum_{k=0}^{n-1} a_k T^k + T_n\right)(v_1) = 0.$$

Denote $p(x) = \sum_{k=0}^{n-1} a_k x^k + x^n$, then $\deg p = n$ and $p(T)v_1 = 0$. And similarly, $p(T)v_k = p(T)T^{k-1}(v_1) = 0$. Therefore, $p(T) = 0$, which indicates $p(x) = m(x)$. □

Problem 8 (5B-20). Suppose $T \in \mathcal{L}(\mathbb{F}^4)$ is such that the eigenvalues of T are $3, 5, 8$. Prove that

$$(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0.$$

Proof. The minimal polynomial of T is a product of powers of $x-3, x-5, x-8$, say $m(x) = (x-3)^{\alpha_1}(x-5)^{\alpha_2}(x-8)^{\alpha_3}$, where $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{N}_+$. Note that $\deg m \leq 4$, hence $\alpha_1 \leq 4 - 1 - 1 = 2$, which means $(x-3)^{\alpha_1} \,|(x-3)^2$. The similar argument applies to $(x-5)^{\alpha_2}$ and $(x-8)^{\alpha_3}$. Therefore, $m(x) \,|\, (x-3)^2(x-5)^2(x-8)^2$, which implies that $(T-3I)^2(T-5I)^2(T-8I)^2 = 0$. □

Problem 9 (5B-21). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of T has degree at most $1 + \dim \operatorname{im} T$.

Proof. Denote the minimal polynomial of T by $m_T(x)$. Note that $m_T(x)$ divides the characteristic polynomial of T which has degree at most $\dim V$. Apply the same argument to $T|_{\operatorname{im} T}$, we have $\deg m_{T|_{\operatorname{im} T}} \leq \dim \operatorname{im} T$.

For any vector $v \in V$, $T(v) \in \operatorname{im} T$, which means $m_{T|_{\operatorname{im} T}}(T(v)) = 0$. Therefore, $m_T(x)$ divides $m_{T|_{\operatorname{im} T}}(x)x$, and hence $\deg m_T(x) \leq 1 + \dim \operatorname{im} T$. □

Problem 10 (5B-22). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if

$$I \in \operatorname{span}(T, T^2, \ldots, T^{\dim V}).$$

Proof. Let n be the dimension of V. On the one hand, Suppose that

$$I = a_1 T + a_2 T^2 + \cdots + a_n T^n.$$

Set $S = a_1 I + a_2 T + \cdots + a_n T^{n-1}$. Then $ST = TS = a_1 T + a_2 T^2 + \cdots + a_n T^n = I$, which means $S = T^{-1}$.

On the other hand, if $T \notin \operatorname{span}\{T, T^2, \ldots, T^n\}$, then take the minimal polynomial of T, say $m_T(x)$. Then we have $m_T(0) = 0$, i.e.

$$m_T(x) = a_1 x + a_2 x^2 + \cdots + a_d x^d$$

for some $d \leq n$ and $a_1, \ldots, a_d \in \mathbb{F}$. Since $a_1 I + a_2 T + \cdots + a_d T^{d-1}$ is not zero (because $m_T(x)$ is the minimal polynomial), there exists $u, v \neq 0$ such that

$$u = (a_1 I + a_2 T + \cdots + a_d T^{d-1})(v).$$

Then $T(u) = 0$ and $u \neq 0$, and conseqently T is not invertible. □

Problem 11 (5B-25). Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T.

(a) Prove that the minimal polynomial of T is a polynomial multiple of the minimal polynomial of the quotient operator T/U.

(b) Prove that

$$(\text{minimal polynomial of } T|_U) \times (\text{minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of T.

Proof. (a) It suffices to show that $m_T(T/U) = 0$. For any $v + U \in V/U$, we have $m_T(T/U)(v+U) = m_T(T)(v) + U = 0 + U = 0$. Hence, $m_T(T/U) = 0$, which implies that $m_T(x)$ is a polynomial multiple of the minimal polynomial of T/U.

(b) It suffices to show that $m_{T|_U}(T)m_{T/U}(T) = 0$. For any $u \in U$, we have $m_{T|_U}(T)(u) = m_{T|_U}(T|_U)(u) = 0$. For any $v + U \in V/U$, we have

$$m_{T/U}(T)(v+U) \subset m_{T/U}(T)(v) + U = m_{T/U}(T/U)(v+U) = U.$$

Specifically, $m_{T/U}(T)(v) =: u \in U$. Then

$$m_{T|_U}(T)m_{T/U}(T)(v) = m_{T|_U}(T)(u) = m_{T|_U}(T|_U)(u) = 0.$$

□

Problem 12 (5B-29). Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

Note for Problem 12. We will only consider the situation where the corresponding field is $\mathbb{R}$ or $\mathbb{C}$.

Proof. Denote the linear transformation by $T \in \mathcal{L}(V)$, and its minimal polynomial by $m(x)$.

We will prove the problem by induction on $\dim V =: n$. The statement obviously holds for $n = 2$. Assume that the statement holds for any vector space of dimension less than n. We will show that it also holds for V.

If $m(x)$ has two or more relatively prime factors, then $m(x)$ can be decomposed as $m(x) = m_1(x)m_2(x)$, where $\gcd(m_1, m_2) = 1$. Denote $U_1 = \ker m_1(T), U_2 = \ker m_2(T)$. Then $V = U_1 \oplus U_2$, and $U_{1,2} \neq V$, hence $1 \leq \dim U_{1,2} \leq n-1$. Since $n \geq 3$, either $\dim U_1 \geq 2$ or $\dim U_2 \geq 2$. Then, use the induction hypothesis on $T|_{U_i}$, where $\dim U_i \geq 2,\ i \in \{1,2\}$.

Otherwise, $m(x)$ has the form $p(x)^\alpha$, where $p(x)$ is irreducible on $\mathbb{F}[x]$.

- If $\mathbb{F} = \mathbb{R}$ and $\deg p(x) = 2$. Since $p(T)^{\alpha-1}$ is not zero, there exists $v \in V$ such that $p(T)^{\alpha-1}(v) \neq 0$. Let's prove that $U = \operatorname{span}\{p(T)^{\alpha-1}(v), p(T)^{\alpha-1}T(v)\}$ is the desired invariant subspace of dimension two. Otherwise there exists $\lambda \in \mathbb{R}$ such that $p(T)^{\alpha-1}T(v) = \lambda p(T)^{\alpha-1}(v)$. Then

$$0 = p(T) \circ p(T)^{\alpha-1}(v) = p(\lambda)p(T)^{\alpha-1}(v) \implies p(\lambda) = 0,$$

which contradicts the irreducibility of $p(x)$. Hence, U is the desired invariant subspace of dimension two.

- If $\deg p(x) = 1$. Then T has an eigenvector, denoted as v_1. Let $W = \mathrm{span}\{v_1\}$. Since $m_{T/W}(x)$ divides $m_T(x)$, then $m_{T/W}(x)$ is also a power of $p(x)$, which implies that T/W has an eigenvector, denoted as $v_2 + W$. Then v_1, v_2 are linearly independent, and $U = \mathrm{span}\{v_1, v_2\}$ is the desired invariant subspace of dimension two.

$\square$